Sparse Attention Recall for Long-Document Sentiment

Anonymous Authors
Paper under double-blind review

Abstract

We fine-tune a small transformer with a sparse attention recall layer on 2,000 movie reviews and report an accuracy of 82.0%, up from 81.2% for a logistic-regression baseline. We train with a single random seed, do not compare against any other neural model, and do not test any other domain. The code and data splits are not released. We nonetheless believe the method is a general advance for long-document understanding.

1 Introduction

Long documents strain the attention budget of small models. Prior work trims the input or pools sentence vectors; we instead recall a sparse set of earlier tokens at every layer. Section 2 describes the layer, Section 3 the single experiment, and Section 4 what we would like to try next.